

Cyber Security: Educational Impact

Caden Kelly • OT Cyber Security Training Modules

ACEP Undergraduate Summer Internship



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Agenda

August 2026

1. Problem Statement/Purpose
2. Research Project Goals
3. The Bigger Picture
4. My Project
5. Challenges and Lessons Learned
6. Conclusions



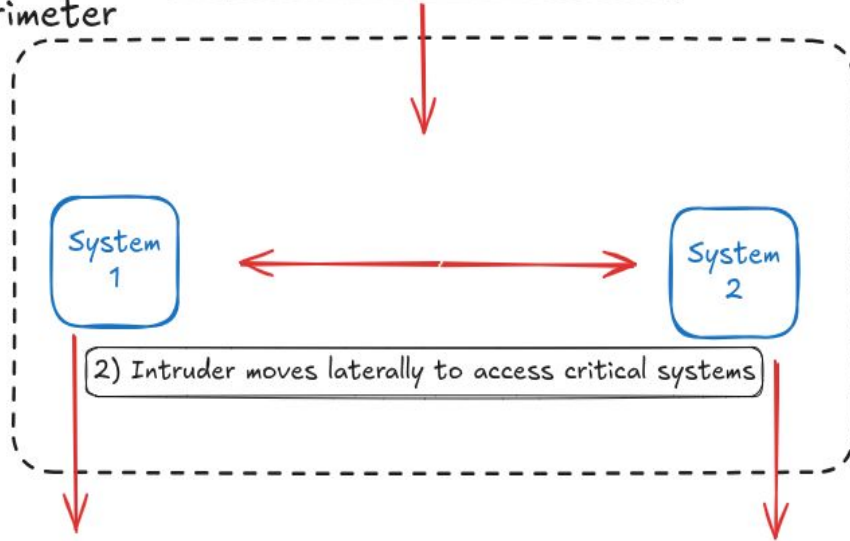
Problem Statement

- Rural Alaskan microgrids often lack modern cyber infrastructure and security features
 - **Security:**
 - **Aging firewalls/single subnets**
 - Resource constraints
 - Limited staff
 - Access to training resources
 - Cost effective solutions

Flat Networks (same subnet)

Security
Perimeter

1) Intruder breaches perimeter firewall

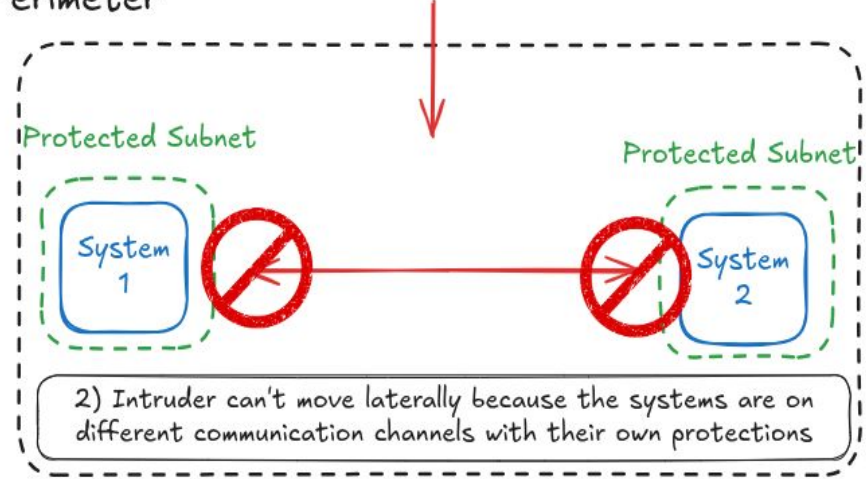


3) Sensitive data and assets are compromised

Segmented Networks

Security
Perimeter

1) Intruder breaches perimeter firewall



2) Intruder can't move laterally because the systems are on different communication channels with their own protections



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Problem Statement

- Rural Alaskan microgrids often lack modern cyber infrastructure that lacks modern cyber security features
 - Security:
 - Aging firewalls/single subnets
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Research Project Goals

OT Cyber Security Training Modules

- Demonstrate a cost-effective segmented network
- Low-cost training for beginners
- Portability compounds effects
- Modular - future potential



The Bigger Picture

ACEP Goals:

Mission

“To develop and disseminate effective, economic and innovative energy solutions for Alaska and beyond.”

Vision

“Alaska leading the way in innovative, secure and resilient energy systems.”

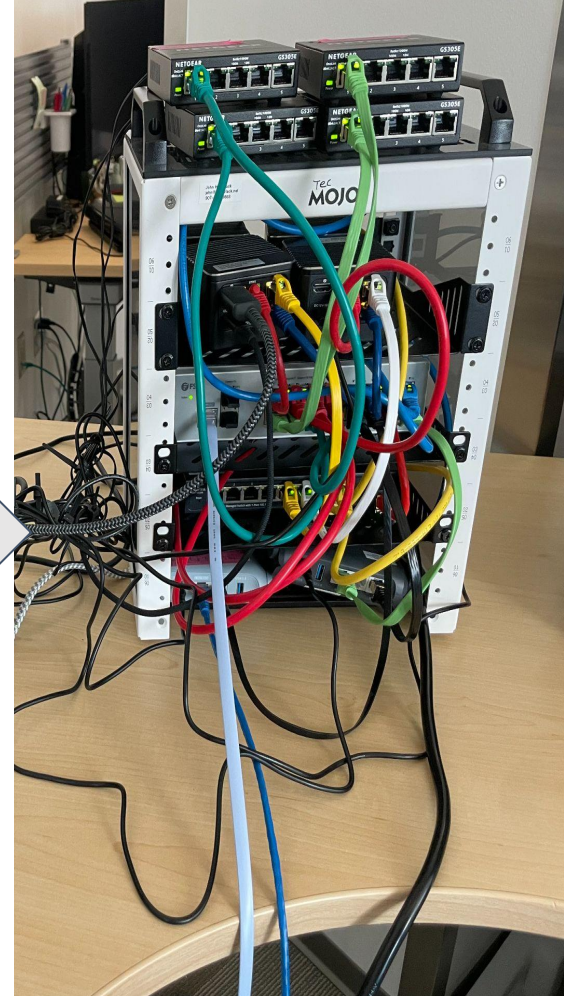
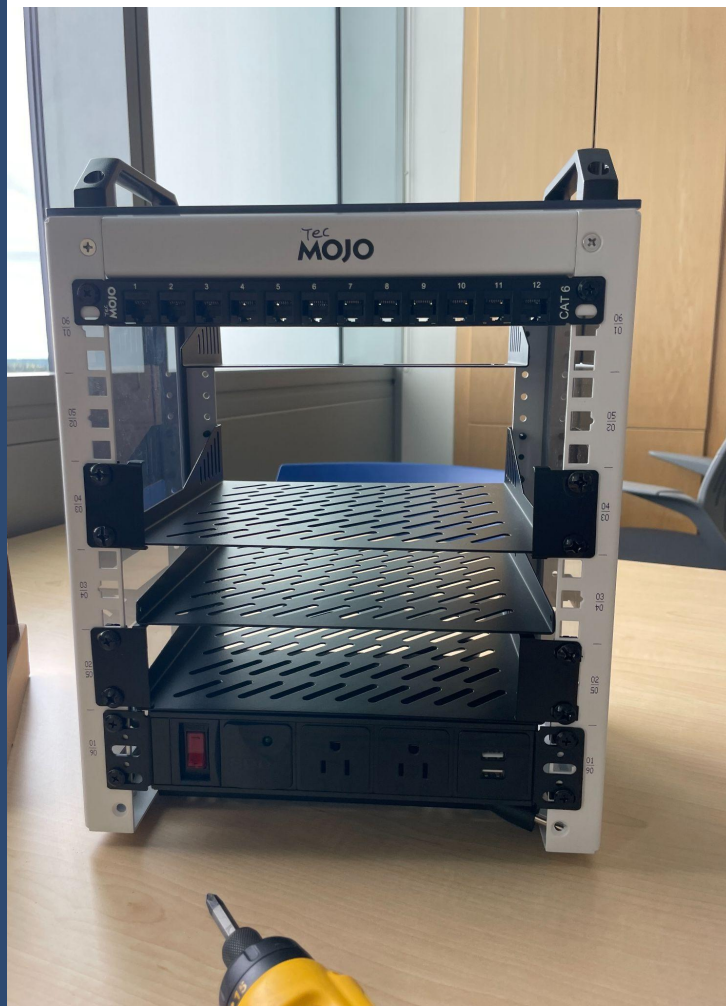


My Project

- 1) Orientation
- 2) **Building Portable Training Ground**
- 3) Training Module Creation
- 4) Testing



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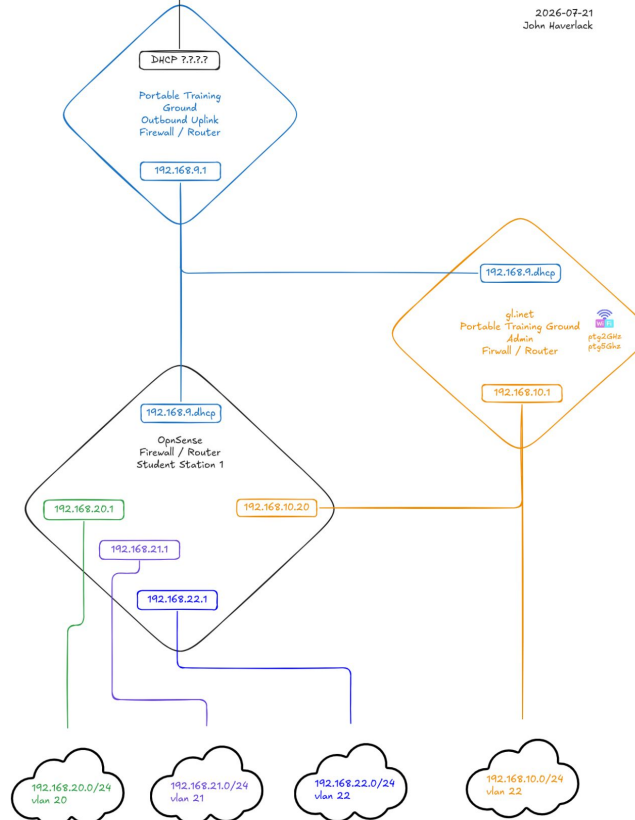
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Portable Training Ground
Student Station
Layer 3 Network Diagram

2026-07-21
John Haverlack



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Not securehttps://192.168.10.10:8006/#v1.0:-node%2Ftg-pve-3:4:5:::5

PROXMOX Virtual Environment 9.2.3

DocumentationCreate VMCreate CTroot@pam

Server View

Node 'tg-pve-3'

RebootShutdownShellBulk ActionsHelp

Server View

Node 'tg-pve-3'

RebootShutdownShellBulk ActionsHelp

Package versions

tg-pve-3 (Uptime: 02:37:38)

CPU usage0.00% of 8 CPUs

Load average2.85,2.42,1.88

IO delay0.00%

RAM usage18.04% (2.77 GiB of 15.36 GiB)

KSM sharing0 B

HD space25.68% (9.92 GiB of 38.63 GiB)

SWAP usage0.00% (0 B of 8.00 GiB)

CPU(s)8 x Intel(R) Core(TM) i3-N305 (1 Socket)

Kernel VersionLinux 7.0.6-2-pve (2026-05-29T11:08Z)

Boot ModeEFI

Manager Versionpve-manager/9.2.3/d0fde103346cf89a

Repository StatusProxmox VE updates Non production-ready repository enabled

Server Load

Load average

Memory usage

TotalUsedZFS ARCAvailable

Search

Summary

Notes

Shell

System

Network

Certificates

DNS

Hosts

Options

Time

System Log

Updates

Repositories

Firewall

Disks

LVM

LVM-Thin

Directory

ZFS

tg-pve-0

tg-pve-1

tg-pve-2

tg-pve-3

Firewall

dhcp-admin

ptg-pool-all

ptg-pool-s1

ptg-pool-s2

ptg-pool-s3

ptg-pool-s4

tg-group-1

tg-group-2

tg-group-3

tg-group-4

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PROXMOX Virtual Environment 9.2.3

Search

Documentation Create VM Create CT root@pam

Server View

Datacenter (PTG)

tg-pve-0

- 101 (ptg-V20-Static)
- 102 (ptg-V20-M1)
- 105 (ptg-V20-M1)
- 120 (ptg-swm2-20)
- 121 (ptg-swm2-21)
- 122 (ptg-swm2-22)
- 600 (V20Student1-OPNsense)
- 998 (ptg-OPNsense)
- 999 (ptg-dhcp)
- 100 (ptg-V20-M1)
- 103 (deb13-impl-260610)
- localnetwork (tg-pve-0)
 - Backups (tg-pve-0)
 - ceph-vm (tg-pve-0)
 - local (tg-pve-0)
 - local-vm (tg-pve-0)
- tg-pve-1
 - 106 (ptg-V20-M1)
 - 107 (ptg-V20-M1)
 - 108 (ptg-V20-M1)
 - 130 (ptg-swm3-30)
 - 131 (ptg-swm3-31)
 - 132 (ptg-swm3-32)
 - 604 (V30Student2-OPNsense)
 - 997 (ptg-OPNsense)
 - localnetwork (tg-pve-1)
 - Backups (tg-pve-1)

Q Search

Summary

Notes

Cluster

Ceph

Options

Storage

Backup

Replication

Permissions

Users

API Tokens

Two Factor

Groups

Pools

Roles

Realms

HA

SDN

Zones

Datacenter

Health

Status

Nodes

Ceph

Cluster: PTG, Quorate: Yes

HEALTH_WARN

Resources

CPU

Memory

Storage

2% of 32 CPU(s)

25% 15.53 GiB of 61.42 GiB

18% 316.63 GiB of 1.74 TiB

Subscriptions

No Subscription

Guests

Virtual Machines

LXC Container

Running 2

Stopped 32

Templates 2

Running 0

Stopped 0

Nodes

N...	ID	Online	Support	Server Address	CPU usage	Memory usage	Uptime
tg-...	1	✓	-	192.168.10.10	8%	24%	2 days 22:...
tg-...	2	✓	-	192.168.10.11	1%	25%	2 days 22:...
tg-...	3	✓	-	192.168.10.12	1%	28%	2 days 22:...
tg-...	4	✓	-	192.168.10.13	1%	24%	2 days 22:...

My Project

- 1) Orientation
- 2) Building Portable Training Ground
- 3) Training Module Creation**
- 4) Testing

Module 1: IPv4 Network Basics

- M1:
 1. **1:30pm – 1:50pm** - *Your terminal, Linux*
 2. **1:50pm – 2:20pm** - *Binary, IPv4, (ping, broadcast, subnets and their mask, etc.)*
 3. **2:20pm – 2:50pm** - *DHCP/MAC/DNS*
 4. **2:50pm – 3:00pm** - *10 minute break*
 5. **3:00pm – 3:30pm** - *NTP, LAN, Network Segmentation*
 6. **End of Module 1**



Aside from the pinging activity for network segmentation, there were many others...

We can see the translation and the route being taken in real time

- Type "traceroute google.com"

```
student1@ptg-V20-M1:~$ traceroute google.com
traceroute to google.com (142.250.73.110), 30 hops max, 60 byte packets
 1 192.168.20.9 (192.168.20.9)  0.393 ms  0.371 ms  0.360 ms
 2 console.gl-inet.com (192.168.10.1)  1.607 ms  1.651 ms  1.563 ms
 3 192.168.9.1 (192.168.9.1)  3.169 ms  3.154 ms  3.133 ms
 4 10.25.156.1 (10.25.156.1)  5.506 ms  4.047 ms  4.781 ms
 5 10.5.222.17 (10.5.222.17)  3.422 ms  3.405 ms  3.380 ms
 6 10.5.222.13 (10.5.222.13)  3.356 ms  2.161 ms  2.127 ms
 7 10.5.222.3 (10.5.222.3)  2.098 ms  3.052 ms  2.243 ms
 8 10.5.222.2 (10.5.222.2)  2.227 ms  2.680 ms  2.646 ms
 9 172.16.0.227 (172.16.0.227)  3.222 ms  3.173 ms  3.155 ms
10 137.229.252.161 (137.229.252.161)  3.943 ms  3.917 ms  4.074 ms
11 137.229.255.232 (137.229.255.232)  43.689 ms  43.661 ms  43.638 ms
12 ae17--202.icar-sttl1-1.infra.pnw-gigapop.net (209.124.190.216)  43.616 ms  43.588 ms  43.650 ms
13 lo0--200.icar-sttl1-4.infra.pnw-gigapop.net (209.124.190.11)  43.615 ms  43.589 ms  43.566 ms
14 74.125.51.244 (74.125.51.244)  43.544 ms  43.519 ms  43.498 ms
15 * * *
16 142.250.233.134 (142.250.233.134)  43.747 ms  142.251.50.242 (142.251.50.242)  45.157 ms  142.251.55.202 (142.251.55.202)  43.851 ms
17 pnseaa-an-in-f14.1e100.net (142.250.73.110)  43.830 ms  142.251.64.87 (142.251.64.87)  43.813 ms  pnseaa-an-in-f14.1e100.net (142.250.73.110)  43.799 ms
student1@ptg-V20-M1:~$ _
```


Module 2: Firewalls

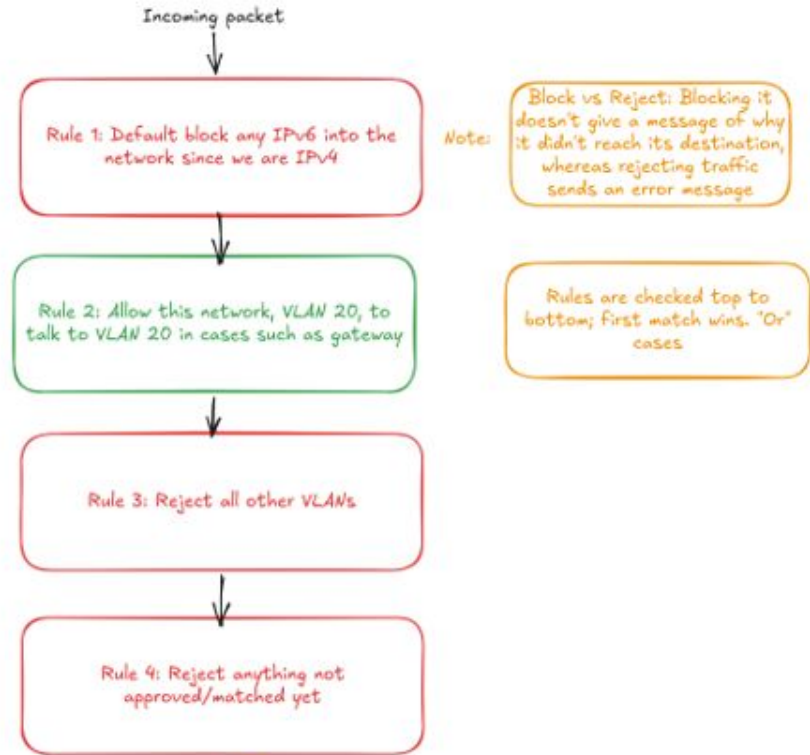
M2:

- **3:30pm – 4:10pm** - *What is a Firewall? Diagram Current Network, Get Into OPNsense WUI*
- **4:10pm – 4:20pm** - *10 minute break*
- **4:20pm – 5:20pm** - *Configure VM DHCP, DNS, and NTP, Testing*
- **5:20pm – 6:20pm** - *Firewall Rules, Configuration, Testing*
- **6:20pm – 6:30pm** - *Feedback Form, END*



Firewall Rules

- A firewall rule is just a statement: **"For this type of traffic, from here, to there, on this port, allow or block it"**
- "Or" analogy: First match wins
- Final "block all else" rule
- Good rules are specific.
- Test your rules immediately



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-  Lobby

 Reporting

 System

 Interfaces

 Firewall

 Aliases

 Categories

 Groups

 NAT

 Rules [new]

 Rules

 Shaper

 Settings

 Log Files

 Diagnostics

 VPN

 Services

 Power

 Help
- 















Firewall: Rules [new]

VLAN20

Categories

Inspect

Tree

 50  

☐		☒	Icons		Version	Protocol	Source	Port	Destination	Port	Gateway	Description	Commands
☐		☒	  		VLAN20	IPv6	*	VLAN20 network	*	*	*	Default deny LAN IPv6 to any rule	    
☐		☒	  		VLAN20	IPv4	*	VLAN20 	*	This Firewall	*	Allow VLAN 20 to talk to VLAN 20 in cases such as gateway	    
☐		☒	  		VLAN20	IPv4	*	TG_VLANs 	*	VLAN20 	*	Reject all other VLANs	    
☐		☒	  		VLAN20	IPv4	*	*	*	*	*	Reject anything not approved/matched yet	    

  1  

Showing 1 to 4 of 4 entries      

Apply

Note: all challenges only need you to edit the VLAN interface for your VM. They will only require you to work in your current groups

Challenge 2:

Goal: Make it so **Teammate B** and **Teammate C** can ping **each other**, in both directions. **Teammate A** should still be unable to reach either of them, and should still be unreachable by both.



Challenges

- Challenges:
 - **Hardware + University limitations leading to unusual setups**
 - Learning EVERY switch interface





Challenges

- Challenges:
 - Hardware + University limitations leading to unusual setups
 - **Learning EVERY switch/router interface**



Lessons Learned

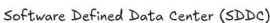
- Advice for picking up where I left off
 - **Understanding, not memorization**
 - Diagrams!!!
 - Backups, snapshots, documentation



Lessons Learned

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 - **Diagrams!!!**
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Lessons Learned

- Advice for picking up where I left off
 - Understanding, not memorization
 - Diagrams!!!
 - **Backups, documentation**

Virtual Machine 600 (V20Student1-OPNsense) on node 'tg-pve-0' m2

Summary Console Hardware Cloud-Init Options Task History Monitor Backup Replication **Snapshots**

Take Snapshot Rollback Edit Remove

Name	RAM	Date/Status
Post-installPre-config	No	2026-07-09 14:55:34
ConfigPreRuleMigrati...	Yes	2026-07-15 16:20:53
PostRuleMigration	Yes	2026-07-15 16:25:44
NOW		

Daily Notes Starting 6.4

- 6.1.26-6.3.26 Notes Recap
- 6.4.26
- 6.5.26
- 6.8.26
- 6.9.26
- 6.10.26
- 6.11.26
- 6.12.26
- 6.15.26
- 6.16.26
- 6.17.26
- 6.18.26: Module Planning Outline Linked...
- 6.19.26
- 6.22.26
- 6.23.26
- 6.24.26
- 6.25.26
- 6.26.26
- 6.29.26
- 6.30.26
- 7.1.26 Training Modules: Slides Linked H...
- 7.6.26
- 7.7.26
- 7.8.26
- 7.9.26
- 7.10.26
- 7.13.26
- 7.14.26
- 7.15.26

7.9.26

Done

- Tag OPNsense VMs
- OS Excalidraw Diagram for students that is simple made and imported (all images used in diagram are from vault's Flat Icon folder in Library-images folders)
- CDR image source attributed in speaker notes
- Linux cheat sheet added noolisp, sudo, traceroute
- add lesson identifiers at bottom of helps
- Set up permissions on Proxmox Accounts with pools for resources
- Make terminal windows more clear in presentation by cutting out what isn't needed and making pics bigger, and taking some words off since I will just be saying it; slides of pictures with speaker notes are okay
- Installed an unconfigured opnsense VM Each student will be able to configure. Added one per student, in the appropriate pools.
 - Made snapshots of each in case students brick themselves out of it

Notes

- Instructor is the tag for the OPNsense VMs I have already set up and will not be having students directly interact with. These will be shut off when students configure their own OPNsense.
 - m1 is tag for module 1 vms they'll log into
 - Did custom tag style from options
- Need to add the opnsense VMs they will be configuring to their pools once they see them
- student's grouprole/pool was useless so I deleted it. Don't have a fourth student per node.



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Limitations

- Amazing mentor gone for 2 weeks straight, + Dr. Checkups
 - Velocity vs. struggle-learning
- Hardware: unideal configurations
- Few testers



Implications

Training effective on
tested groups



Can scale to rural
Alaskan communities

Cost-effective setup



People can practice/
configure on their own



Future Directions

- Modules 3, 4...
 - The training ground I built
 - Additional security practices
 - Etc.
- Portability + Scalability
 - Further testing, deployment to communities, impact data collection
- Different hardware configurations
 - Cabling, reduction of devices, etc.

Conclusion

- Successful trainings as proving ground
- I learned **so much**
- Future directions:
 - Additional modules, testing, deployment
- Enjoy least:
 - Debugging simple mistakes
- Enjoy most:
 - Learning new topics, real skillset, bettering people's' lives



Contact

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Q&A



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